Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

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This document is the complete technical reference for the Neural Field Perceiver project. It unifies the theoretical problem framing, cross-disciplinary foundations from 3D computer vision, the full architecture specification, uncertainty estimation, and the experimental plan. Claims throughout are categorized as **externally established** (published literature), **internally observed** (our experiments), or **working hypotheses** (plausible but unverified).

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Part I

Problem Framing

1 The Physical Problem

Each patient's micro-electrocorticography (μECoG) array is a **sparse spatial sample** of neural activity over the cortical surface during speech production. The array is placed subdurally over left ventral sensorimotor cortex (vSMC) during surgery, and its position in standardized brain space (MNI-152) is determined post-hoc via electrode reconstruction (BioImage Suite for DBS patients, Brainlab for tumor patients).

1.1 The cortical surface, not free 3D space

The cortex is a folded two-dimensional sheet, not a volume. Let $\mathcal{S}$ denote the cortical surface embedded in $\mathbb{R}^3$. The high-gamma analytic amplitude (HGA, 70–150 Hz envelope) is a function on this surface:

$$a : \mathcal{S} \times \mathbb{R} \to \mathbb{R} \tag{1}$$

For patient i with N_i electrodes at cortical surface positions $\{\mathbf{p}_j^{(i)}\}_{j=1}^{N_i} \subset \mathcal{S}$, we observe:

$$\boxed{d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t)} \tag{2}$$

where ϵ captures measurement noise, electrode impedance variation, and sub-electrode-scale neural activity not resolved by the $\sim$2 mm contact spacing.

In practice, we approximate cortical surface positions with 3D MNI coordinates: $\mathbf{p}_j^{(i)} \approx (x_j^{(i)}, y_j^{(i)}, z_j^{(i)})_{\text{MNI}}$. This approximation has two consequences:

1. **Euclidean distance in MNI space does not equal geodesic distance on the cortical surface.** Two electrodes 3 mm apart in MNI may sit on opposite banks of a sulcus, making them functionally distant. For μECoG arrays on exposed cortex, the local patch under the array is approximately planar, so within-array Euclidean distances are reasonable. Cross-patient comparisons at larger scales are more affected by folding geometry.
2. **Surface-based coordinates** (e.g., projection to `fsaverage`) would better respect cortical geometry than volumetric MNI coordinates. Whether surface reconstructions exist for our patients depends on the BioImage Suite / Brainlab outputs available.

The decoding problem. Given observations $\{d_j^{(i)}(t)\}$ at approximately known surface positions, infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$ that produced the spatiotemporal pattern.

The cross-patient problem. Train on patients $\{1, \ldots, K\}$ and decode for a held-out patient $K{+}1$ whose electrodes sample the cortical surface at *different positions*. The observations are not aligned: electrode j on patient i and electrode j on patient i' correspond to different cortical locations.

Factor	Quantity	Implication
Electrodes per patient	63–201	Sparse sampling of the cortical field
Electrode spacing	∼2 mm	Resolves individual gyral columns
Array offset across patients	15–25 mm in MNI	Different cortical regions sampled
Array rotation	Variable	Channel indices meaningless across patients
Correspondence uncertainty	Several mm	Coordinate-based alignment is approximate
Cortical folding variability	Patient-specific	Sulcal geometry differs across individuals
Tumor mass effect (subset)	Variable	Macroscopic warping in 6 of 12 patients
Trials per patient	46–178	Small-data regime for supervised learning
Total utterance per patient	∼1 min	Insufficient for single-patient SSL
Phoneme vocabulary	9 classes × 3 positions	27-way classification at each trial

1.2 What makes this hard

2 The Shared Dynamics Hypothesis

Cross-patient decoding is only possible if the cortical field $a(\mathbf{p}, t)$ shares structure across patients. We decompose:

$$\boxed{a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t)} \tag{3}$$

where:

- $f(\cdot, t; \mathbf{y})$ is the **shared neural field**—the spatiotemporal pattern that phoneme sequence $\mathbf{y}$ produces in canonical cortical space.
- $T_i : \mathcal{S}_i \to \mathcal{S}_{\text{canonical}}$ is a **patient-specific spatial transform** mapping each patient's cortical surface to a canonical reference. MNI normalization is our estimate of T_i, but it is imperfect.
- $r^{(i)}(\mathbf{p}, t)$ is a **patient-specific residual** capturing individual neural organization, signal characteristics, and functional plasticity.

The viability of cross-patient decoding depends on the ratio $\|f\|/\|r^{(i)}\|$ at the spatial resolution accessible to our measurements and alignment procedure.

2.1 Evidence for shared dynamics

2.1.1 Externally established (published literature)

Per-patient input layers + shared temporal backbone is the optimal transfer architecture. Singh et al. (2025) demonstrate group training with N=25 sEEG patients: per-patient Conv1D + shared BiLSTM (2×64), frozen shared backbone + fine-tuned per-patient layers yields PER 0.49 vs. 0.57 single-subject (p<0.001). Willett et al. (2023) use per-day affine input layer (256→256) + shared 5-layer GRU (512 hidden) for day-to-day BCI transfer.

Caveat: Singh's data regime is ∼60 min/patient (vs. our ∼1 min); Conv1D hyperparameters are unreported. Willett is N=1 with ∼40 min/day. The architectural pattern transfers; the hyperparameters do not.

Articulatory somatotopy in vSMC is reproducible within individuals. Metzger et al. (2023) show somatotopic organization of speech articulators in one patient with chronic ECoG (253 channels): labial, front tongue, back tongue, and vocalic representations are spatially organized

(p<0.0001), preserved even after 18 years of anarthria, suggesting the organization is intrinsic to motor cortex circuitry.

Caveat: N=1 chronic implant. Cross-patient spatial consistency is inferred from anatomical uniformity but not directly measured in any paper in our corpus.

Articulatory features are a robust encoding basis in vSMC. Wu et al. (2025) reconstruct speaker-invariant articulatory features from HD-ECoG over vSMC (PCC 0.75–0.80, N=3), including one intra-operative patient (PCC = 0.76). The claim that articulatory features are "more robustly encoded than acoustic features" is attributed by Wu to Chartier (2018), Conant (2018), and Anumanchipalli (2019).

MNI coordinates enable cross-patient ECoG transfer. Chen et al. (2025, SwinTW) use MNI coordinates and ROI embeddings as positional information across 43 ECoG + 9 sEEG subjects, reporting mean PCC = 0.765 on unseen participants in cross-validation.

Caveat: Standard ECoG has ∼10 mm spacing, comparable to inter-subject correspondence uncertainty. Our μECoG spacing (∼2 mm) is finer, meaning coordinates may encode global array position but not resolve within-array relationships.

Motor cortex neural manifold structure is preserved across individuals. Safaie et al. (2023, *Nature Neuroscience*) show that motor cortex population dynamics during reaching are geometrically similar across different monkeys. Separately, Gallego et al. (2020, *Nature Neuroscience*) establish that low-dimensional manifold dynamics are preserved within individuals across long time periods, even as recorded neurons change—evidence for temporal stability of latent dynamics.

Caveat: Safaie concerns limb motor cortex in non-human primates. Gallego is within-individual. The extension to cross-individual speech motor cortex is plausible but untested.

Consistent articulatory somatotopy across participants. Bouchard et al. (2013, *Nature*) used high-density ECoG to demonstrate reproducible articulatory maps in vSMC across 3 participants. Anumanchipalli et al. (2019, *Nature*) reconstructed continuous articulatory trajectories from motor cortex.

2.1.2 Internally observed (our experiments)

Cross-patient supervised transfer works at this data scale. Our LOPO pipeline (per-patient SpatialConv + shared BiGRU backbone, 9 source patients) achieves PER = 0.762 on held-out S14, compared to per-patient PER = 0.737 (grouped-by-token CV). This is direct evidence that decodable shared structure exists in f.

Articulatory bottleneck head improves LOPO decoding. Projecting to a 15-dimensional phonological feature space via a fixed linguistic matrix outperforms a flat CE head for cross-patient decoding, consistent with f being organized along articulatory feature dimensions.

55 experiments on S14 converge to PER 0.750–0.780. Across supervised, SSL, domain adaptation (DANN, CORAL, CCA), self-training, and knowledge distillation paradigms. See Section 5.4 for interpretation.

SSL features carry partial structure. JEPA features + LP-FT achieve PER = 0.797 (vs. 0.854 with basic evaluation). Supervised features with the same recipe achieve 0.737. All SSL methods fail at current data scale but feature geometry is non-trivial.

2.1.3 Working hypotheses (plausible but unverified)

The ∼0.76 PER convergence on S14 reflects (in part) a spatial alignment bottleneck. This has not been isolated from target-data scarcity, measurement noise, or S14-specific effects.

MNI coordinate-based spatial alignment will improve decoding. The practical gain depends on registration accuracy, model architecture, and whether spatial alignment is binding.

Cross-task data pooling (Duraivel 2025 pseudoword repetition, same 9 phonemes, 3 overlapping patients) is a high-value, low-risk path to more data.

3 The Neural Manifold Connection

3.1 Two views of the same phenomenon

The cortical field model (Section 2) and the neural manifold hypothesis describe the same system at different levels:

- **The field view** (spatial): Activity $a(\mathbf{p}, t)$ is a function on the cortical surface. Each electrode samples this field at a known position.
- **The manifold view** (population): The vector of all electrode activities $\mathbf{d}(t) = [d_1(t), \ldots, d_N(t)]$ lies on a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$. Motor behavior corresponds to trajectories on $\mathcal{M}$.

The bridge is a **latent state variable**. Let $\mathbf{z}(t) \in \mathbb{R}^k$ be the low-dimensional articulatory state—the configuration of the vocal tract at time t. Each electrode's activity is determined by *where it sits* on the cortical surface and *what the articulatory state is*:

$$\boxed{d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + \epsilon_j^{(i)}(t)} \tag{4}$$

where $g : \mathcal{S} \times \mathbb{R}^k \to \mathbb{R}$ is a **shared observation function** mapping (cortical position, articulatory state) to neural activity. This is the somatotopic map made explicit: electrodes over lip motor cortex respond to lip-state variables; electrodes over tongue motor cortex respond to tongue-state variables.

The dynamics on the manifold are determined by the phoneme sequence:

$$\dot{\mathbf{z}}(t) = h\big(\mathbf{z}(t);\ \mathbf{y}\big) \tag{5}$$

3.2 The factorization is the key insight

Combining the shared dynamics hypothesis with the manifold hypothesis: the shared field $f(\mathbf{p}, t; \mathbf{y})$ factorizes as $g(\mathbf{p}, \mathbf{z}(t))$, separating space from time through a low-dimensional bottleneck. Instead of learning an arbitrary spatiotemporal function, the model learns:

1. A spatial function g: how each brain position encodes articulatory features (the somatotopic map).
2. A dynamical system h: how the articulatory state evolves during speech.

Contrast	Phonemes distinguished	Dim.
Place of articulation	labial (b,p,v) vs. velar (g,k)	∼2
Manner	stop vs. fricative vs. vowel	∼2
Voicing	voiced (b,g,v) vs. voiceless (k,p)	∼1
Vowel height	low (a,æ) vs. high (i,u)	∼1
Vowel backness	front (æ,i) vs. back (a,u)	∼1
Effective manifold dimension		**∼5–8**

3.3 Articulatory manifold dimensionality

The manifold has physical meaning corresponding to vocal tract configuration. For our 9 phonemes, the contrasts that distinguish the inventory:

This is far below the electrode count (63–201), consistent with massive oversampling of a low-dimensional manifold. The decoding problem reduces to: given sparse samples of $g(\mathbf{p}, \mathbf{z})$ at known positions $\mathbf{p}$, infer the latent trajectory $\mathbf{z}(t)$ and classify the phoneme sequence.

3.4 Cross-patient transfer as manifold alignment

Each patient's electrode array gives a different **projection** of the shared manifold:

$$\mathbf{d}^{(i)}(t) \approx W^{(i)}\mathbf{z}(t) + \boldsymbol{\epsilon}^{(i)} \tag{6}$$

where $W^{(i)} \in \mathbb{R}^{N_i \times k}$ is determined by which cortical positions the electrodes sample. Different patients have different $W^{(i)}$ because they have different electrode positions—different projections of the same underlying manifold.

Existing alignment methods (CCA, Procrustes, CORAL) attempt to align these projections post-hoc. All failed in our experiments. MNI coordinates offer an alternative: **aligning the projections a priori** by providing the geometric relationship between electrode positions.

The key question is whether inter-subject correspondence uncertainty (several mm) is small enough relative to the somatotopic map scale (∼15–20 mm) that MNI coordinates provide useful alignment.

4 Decomposing Cross-Patient Variation

4.1 The spatial transform T_i

The primary source of cross-patient variation is **geometric**: different electrode positions sample different regions of the shared field. The transform T_i decomposes:

$$T_i(\mathbf{p}) = A_i \cdot \mathbf{p} + \mathbf{b}_i + \delta_i(\mathbf{p}) \tag{7}$$

- $A_i \in \mathbb{R}^{3\times 3}$: rigid or affine transform (rotation, scaling). MNI normalization estimates this.
- $\mathbf{b}_i \in \mathbb{R}^3$: translation (15–25 mm array placement offset).
- $\delta_i(\mathbf{p})$: nonlinear residual—sulcal geometry, local folding, and for tumor patients, **macroscopic warping from mass effect**. Tumor displacement is a geometric mapping error, not signal noise: it physically displaces the somatotopic map relative to skull-based coordinates.

4.2 Inter-subject correspondence uncertainty

The commonly cited "∼5 mm" is not a stable physical constant. The actual uncertainty has multiple sources:

1. **Native-space electrode localization**: Sub-millimeter accuracy—relatively precise.
2. **Brain shift correction**: Intra-operative brain shift (craniotomy, CSF drainage, gravity). Correction quality varies.
3. **Warping to common atlas space**: The dominant source. Quality depends on algorithm, template, and lesion distortion.

Net result: **several millimeters** for healthy anatomy, potentially more near tumors:

- At the scale of articulatory somatotopy (∼15–20 mm zones): correspondence is reliable.
- At the scale of electrode spacing (∼2 mm): correspondence is unreliable.
- In between (∼5–10 mm): partially informative, varies by patient and region.

4.3 Resolution mismatch implies dual spatial encoding

μECoG electrode spacing (∼2 mm) is finer than inter-subject correspondence uncertainty. Any coordinate-aware architecture should therefore use **dual spatial encoding**:

Scale	Resolution	Best encoded by	What it captures
Cross-patient	∼10–15 mm	MNI coordinates	Articulatory somatotopy
Intra-patient global	∼5–10 mm	MNI (approximate)	Array position on cortex
Intra-patient local	∼2 mm	Grid topology	Fine-grained spatial patterns

4.4 The patient-specific residual $r^{(i)}$

After accounting for shared dynamics and spatial transform, the residual captures:

1. **Fine-grained neural organization**: Within-articulator spatial patterns varying across individuals.
2. **Signal characteristics**: Impedance, contact quality, SNR. Z-score normalization addresses amplitude scaling but not higher-order distributional differences.
3. **Functional plasticity**: Speech development, accent, articulatory habits producing patient-specific tuning.
4. **Pathology-specific effects**: Parkinson's motor cortex excitability changes; tumor-adjacent cortical reorganization (distinct from δ_i).

5 What Our Experiments Reveal

5.1 The evaluation recipe dominates at this data scale

Across experiments on S14, training improvements contributed ∼7 pp and evaluation improvements contributed ∼6.5 pp:

Component	Category	PER improvement
Label smoothing 0.1	Training	−4.1 pp
Mixup α=0.2	Training	−1.7 pp
Per-position heads + dropout	Training	−1.2 pp
Weighted k-NN (k=10, cosine)	Evaluation	−2.1 pp
TTA (16 augmented copies)	Evaluation	−1.4 pp
Articulatory + CE dual head	Evaluation	−1.1 pp
Best-of per fold	Evaluation	−1.8 pp

This suggests the backbone already extracts most of the signal at this data scale; the bottleneck is extracting predictions from the representation.

5.2 Multi-scale temporal structure is validated

Multi-scale temporal encoding (stride 3+5+10) was the single best architectural improvement (PER 0.766 → 0.750):

Scale	Temporal resolution	Articulatory correlate
Stride 3 (∼67 Hz)	∼15 ms	Fast transitions (stop release, tongue flap)
Stride 5 (∼40 Hz)	∼25 ms	Syllable dynamics (CV transition)
Stride 10 (∼20 Hz)	∼50 ms	Phoneme-level patterns (steady-state)

5.3 SSL features carry partial structure

All SSL methods produce near-chance PER (0.85–0.91) under basic evaluation. JEPA features + LP-FT achieve PER = 0.797, and weighted k-NN improves by 2.9 pp. SSL features capture some structure but are far weaker than supervised features. TTA hurts JEPA features (−0.2 pp), confirming they are not augmentation-invariant.

5.4 The ∼0.76 PER convergence on S14

55 experiments across all paradigms converge to PER 0.750–0.780 on S14. **Attributing this to any single bottleneck would be premature**:

1. **S14 is no longer held-out.** With 55+ architectures evaluated, it functions as a validation set. The convergence may reflect S14-specific characteristics.
2. **Measurement noise from small validation folds.** ∼30 samples per fold implies ±0.05 PER noise. Differences <2 pp are within noise.
3. **Target-data scarcity.** S14 has 153 trials. Stage 2 adaptation contributes only ∼1.6 pp.
4. **Spatial alignment may be one factor.** Per-patient spatial encoders cannot share spatial knowledge. But this has not been isolated as *the* bottleneck.

Breaking through requires: **(a)** population-level evaluation on all patients, **(b)** cross-task data pooling, and **(c)** MNI coordinate-based spatial alignment. These are complementary, not competing.

6 The Case for MNI Coordinates

6.1 Currently unused information

The current LOPO pipeline processes spatial information through **per-patient Conv2d** operating on grid topology:

$$d^{(i)} \in \mathbb{R}^{H_i \times W_i \times T} \xrightarrow{\text{Conv2d}_i} \mathbb{R}^{d \times T} \xrightarrow{\text{shared backbone}} \mathbb{R}^{d' \times T'}$$

Each patient's Conv2d is independent. The model has **zero information** about the spatial relationship between electrodes across patients. The shared backbone implicitly assumes its input features are homologous across the channel dimension—that feature dimension k from different patients' SpatialConv layers carries corresponding information. This assumption is anatomically false: independent per-patient Conv2d layers produce feature spaces with no guaranteed alignment.

6.2 What coordinates enable

The observation function formulation (4) makes the value explicit: knowing $\mathbf{p}_j^{(i)}$ allows the model to learn g as a shared function. Concretely:

1. **Identifying which articulatory zone each electrode samples.** The somatotopic map operates at ∼15–20 mm scale, above correspondence uncertainty.
2. **Shared spatial representations across patients.** Electrodes at similar MNI coordinates can share representations.
3. **Contextualizing variable electrode counts.** Coordinates tell the model which regions are covered and which are absent.

6.3 What coordinates do NOT guarantee

Coordinates help only if: correspondence uncertainty is small enough relative to the somatotopic scale; the model architecture can exploit them; spatial alignment is actually a binding constraint; and the coordinates are accurate enough for specific patients (tumor patients may have larger errors). **MNI coordinates are necessary but not sufficient.** Their value must be demonstrated empirically.

Part II

Cross-Disciplinary Foundations

7 The Multi-View Reconstruction Analogy

The cross-patient decoding problem has a precise structural analog in computer vision: **fitting a parametric 3D model to 2D observations from multiple viewpoints**. This analogy, inspired by holistic performance capture (Hewitt et al., 2024, arXiv:2410.11520), motivates the reprojection loss that is central to our architecture.

7.1 The pose estimation mapping

In 3D human pose estimation (e.g., fitting SMPL to 2D keypoints), a parametric 3D model encodes body structure via pose θ and shape β; a camera projects 3D vertices to 2D via known projection π; sparse, noisy 2D keypoint detections are observed; and the reprojection loss $\sum_j \|x_j - \pi(V_j(\theta, \beta))\|^2$ drives optimization.

3D Pose Estimation		**Neural Field Decoding**
3D body model (SMPL mesh)	$\leftrightarrow$	Shared neural field $g(\mathbf{p}, \mathbf{z})$ on cortex
Pose parameters θ	$\leftrightarrow$	Articulatory state $\mathbf{z}(t)$
Shape parameters β	$\leftrightarrow$	Patient embedding $e^{(i)}$
Camera extrinsics $[R \mid t]$	$\leftrightarrow$	MNI offset $\Delta^{(i)}$
Camera intrinsics K	$\leftrightarrow$	Per-patient gain/bias
2D keypoints x_j	$\leftrightarrow$	Electrode observations $d_j^{(i)}(t)$
Projection function $\pi(\cdot)$	$\leftrightarrow$	Observation function $g(\cdot)$
Keypoint visibility/occlusion	$\leftrightarrow$	Dead/masked electrodes
Reprojection loss $\|\pi(V) - x\|^2$	$\leftrightarrow$	**Reprojection loss** $\|g(\mathbf{p}+\Delta, \hat{\mathbf{z}}) - d\|^2$

7.2 Multi-patient training as multi-view reconstruction

The analogy deepens: multiple patients observing the same cortical dynamics from different electrode positions is **multi-view 3D reconstruction**. 3D structure is shared; only viewpoints differ. Joint optimization recovers structure and camera parameters simultaneously. More cameras (patients) yield better reconstruction. Novel view synthesis becomes novel patient prediction.

7.3 The NeRF connection

Our observation decoder is a **cortical neural field**—structurally analogous to a Neural Radiance Field (NeRF). NeRF takes (x, y, z, θ, ϕ) and outputs (color, density); we take $(\mathbf{p}_{\text{MNI}}, \mathbf{z}_{\text{articulatory}})$ and output HGA activity. NeRF trains by rendering from known cameras; we train by "rendering" at known electrode positions. The key difference: NeRF uses an unconstrained MLP, while our bilinear factorization $\hat{d} = \phi(\mathbf{p})^\top \psi(\mathbf{z})$ is a **structured** neural field, physically motivated by the separability of somatotopic organization and articulatory dynamics.

7.4 What the analogy reveals

Why previous cross-patient methods failed:

- **CCA/Procrustes/CORAL**: Aligning camera views by 2D feature statistics. Works for similar viewpoints but fails for very different ones (15–25 mm offsets = cameras at very different positions).
- **Per-patient SpatialConv**: A separate 3D model per camera—no information sharing.
- **Our approach**: Structure from Motion—jointly estimate shared 3D structure and all camera parameters from multi-view observations. MNI coordinates are approximate GPS for each camera.

Why reprojection loss beats arbitrary SSL: JEPA, VICReg, and masked reconstruction used reconstruction targets with no physical meaning. Reprojection is physically grounded: "if your

latent state estimate is correct, you should predict what every electrode observes at its known brain position." It is self-supervised but geometrically constrained.

8 Validated Patterns from 3D Vision (2024–2026)

8.1 The field's trajectory: per-scene → feed-forward → foundation model

Our architecture sits at step 2 (feed-forward generalizable). Key papers and transferable patterns:

VGGT (Wang et al., CVPR 2025 Best Paper). Single transformer predicts ALL 3D attributes from 1–100+ views in <1s. *Transfer*: Joint prediction of multiple geometric quantities from variable-length observation sets validates our Perceiver jointly estimating latent state + spatial alignment.

pixelSplat / MVSplat (Charatan et al., CVPR 2024 / Chen et al., ECCV 2024). Feed-forward 3DGS from image pairs. MVSplat uses cost volumes for geometry—the analog of our MNI distance-biased attention.

LVSM (Jin et al., ICLR 2025 Oral). Two architectures: encoder-decoder (compresses inputs into fixed latent tokens) vs. decoder-only. Decoder-only wins by 1.5–3.5 dB with enough data. Critical finding: explicit 3D structure is unnecessary at sufficient scale. *For us*: Our virtual electrodes ARE the fixed latent tokens. LVSM validates that at our small data scale, the structured approach (explicit geometry) is correct.

LRM (Hong et al., ICLR 2024). 500M-param transformer: single image → triplane NeRF via cross-attention. Our virtual electrodes serve the same role as the triplane—a fixed canonical sampling of the field.

AnySplat (Jiang et al., SIGGRAPH Asia 2025). Feed-forward from uncalibrated views. Predicts 3DGS + camera parameters simultaneously. Analogous to handling variable electrode count with approximate MNI positions.

8.2 Sparse-view reconstruction

Our setting—63–201 electrodes sampling a cortical field—is analogous to few-view 3D reconstruction.

SPARF (Truong et al., CVPR 2023). NeRF from sparse and **noisy** poses. Directly relevant: our MNI coordinates are noisy poses; our electrodes are sparse views.

The sparse-view 3DGS literature identifies three dominant strategies: (1) monocular depth priors, (2) diffusion priors for hallucination, (3) cost volumes for multi-view matching. Strategy (1) maps to our Grid PE (local structural prior). Strategy (3) maps to MNI distance-biased attention. Strategy (2) requires massive data and does not apply.

8.3 Continuous field reconstruction from sparse sensors

MMGN (Luo et al., ICLR 2024). Reconstructs continuous physical fields (sea surface temperature, climate) from **sparse, irregularly sampled sensor observations** using implicit neural representations. Factorizes spatiotemporal variability into spatial and temporal components via separation of variables. **The most directly relevant paper.** Our problem is almost identical: sparse sensors (electrodes) at known positions (MNI), continuous field (cortical activity), spatiotemporal factorization (our bilinear decoder $\hat{d} = \phi(\mathbf{p})^\top \psi(\mathbf{z})$).

Senseiver (Janny et al., *Nature Machine Intelligence* 2023). Perceiver IO for field reconstruction from sparse sensors—structurally identical to our spatial encoder architecture.

8.4 Architectural pattern validation

Every major component of the Neural Field Perceiver has a validated analog in the 3D vision literature:

Our component	Same pattern in vision
Perceiver cross-attention (virtual → actual)	Cross-attention from canonical to observation tokens (LRM, LVSM, Laser-NV)
Virtual electrodes at learnable positions	Triplane NeRF / fixed latent tokens (LRM, LVSM)
MNI distance-biased attention	Geometry-aware attention / cost volumes (GraphSplat, MVSplat)
Bilinear decoder $\hat{d} = \phi(\mathbf{p})^\top \psi(\mathbf{z})$	Spatiotemporal field factorization (MMGN)
Reprojection loss	Photometric / reprojection loss (all NeRF/3DGS)
Learnable per-patient Δ (pose correction)	Learnable camera extrinsics (SPARF, AnySplat)
Patient embedding $e^{(i)}$	Per-scene latent code (Auto-decoder NeRF, CodeNeRF)
Electrode masking in reprojection	View dropout / masked pretraining (MAE, masked NeRF)
Multi-scale temporal tokenization	Multi-resolution processing (MViT, multi-scale cost volumes)
Dual spatial encoding (MNI + grid)	Multi-level aggregation (cost volume + monocular depth)
Variable electrode count via attention	Variable view count (FreeSplat, VGGT, AnySplat)
Structured inductive bias for small data	Explicit geometry beats decoder-only at small scale (LVSM)

We are not inventing novel architectural patterns—we are transferring proven patterns to a new domain with adaptations for our data regime.

8.5 What we are correctly NOT doing

Technique	Why wrong for us
Diffusion priors for hallucination	Requires millions of training scenes; we have ∼20 patients
4D Gaussian primitives	Our trials are independent events, not a persistent scene
Foundation model scale (500M+)	∼1300 trials cannot support >200K parameters
Decoder-only architecture	Only works with massive data (LVSM's own finding)
Contrastive SSL pretraining	Failed in our experiments (VICReg, BYOL: near-chance)

8.6 The data scale gap

The biggest disconnect between 3D vision and our problem is scale:

	3D Vision	Our Problem
Training scenes/patients	50K–1M+	∼20 patients
Observations per scene	10–100+ images	63–201 electrodes
Total training data	TB scale	∼240 MB
Parameter budget	50M–500M+	∼200K

This means: we cannot use data-hungry techniques; we **must** use strong geometric inductive biases to compensate; our ∼200K architecture is correctly small; and the reprojection loss is especially valuable because it provides per-frame spatial supervision without phoneme labels.

The field's finding that **explicit geometric structure helps at small scale** (LVSM ablation, SPARF) directly validates our design choice.

9 Uncertainty Estimation

9.1 From binary to continuous confidence

Our pipeline already has a **binary** uncertainty mechanism: the significant channel selection (permutation cluster test) that keeps or drops entire electrodes (weight 0 or 1). Proper uncertainty estimation replaces this with **continuous** weighting: each electrode gets a learned confidence $\in (0, 1)$, edge electrodes and high-impedance contacts are downweighted, and per-position predictions carry confidence scores.

9.2 Two types of uncertainty

Aleatoric (irreducible, data-dependent): "This electrode is inherently noisy due to high impedance or poor contact." Per-electrode, does not decrease with more training data.

Epistemic (reducible, model-dependent): "We haven't seen enough data in this cortical region." Per-position, decreases with more patients/data.

Both matter: aleatoric uncertainty should downweight noisy electrodes in spatial aggregation; epistemic uncertainty should flag predictions in poorly-covered cortical regions as unreliable.

9.3 Recommended uncertainty stack

9.3.1 Tier 1: Minimal implementation, high value

Beta-NLL loss (Seitzer et al., ICLR 2022). Replace the MSE reprojection loss with learned per-electrode variance:

$$\mathcal{L}_{\text{reproj}}^{\text{Beta-NLL}} = \frac{1}{2}\Big(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2}\Big) \cdot \big(\hat{\sigma}_j^2\big)_{\text{detach}}^{\beta} \tag{8}$$

where $\beta = 0.5$ prevents variance explosion. The model learns which electrodes are inherently noisy. **Cost**: 3 lines of code. **Gives**: per-electrode aleatoric uncertainty.

MC Dropout at inference (Gal & Ghahramani 2016). Our architecture has Dropout(0.3). Keep dropout active at inference, run 10–20 forward passes, variance across passes = epistemic uncertainty. 2025 best practice: remove dropout from the final output layer. **Cost**: 10–20× inference (model is small—200K params). **Gives**: per-position epistemic uncertainty.

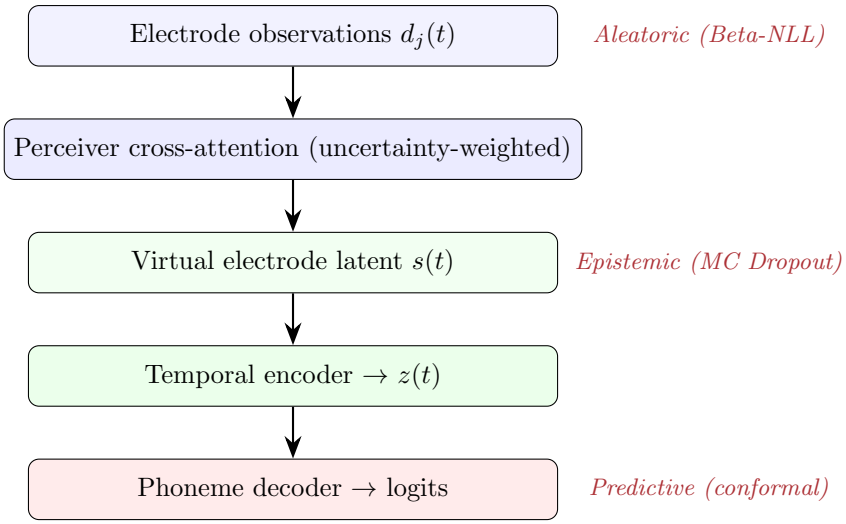
Conformal prediction (Angelopoulos & Bates 2021). Post-hoc, distribution-free: on a calibration set, compute nonconformity scores; at test time, produce prediction **sets** with guaranteed coverage (e.g., "the true phoneme is in this set with 90% probability"). **Cost**: zero training cost; use CV fold as calibration. **Gives**: per-position phoneme prediction sets with statistical coverage guarantees. **Clinical value**: "the model thinks position 2 is /b/ or /p/ (both labial stops)"—interpretable uncertainty.

9.3.2 Tier 2: Moderate implementation, strong grounding

Evidential regression (Amini et al., NeurIPS 2018; Flexible EDL, NeurIPS 2025). Predict Normal-Inverse-Gamma parameters $(\mu, \nu, \alpha, \beta)$ per electrode. Single-pass decomposition: aleatoric $= \beta/(\alpha-1)$, epistemic $= \beta/(\nu(\alpha-1))$. Replaces MC Dropout with single forward pass.

Probabilistic virtual electrodes (pixelSplat-style). Instead of deterministic Perceiver output, predict a **distribution**: $\mu, \log \sigma^2 = \text{split}(\text{cross_attention_output})$. Virtual electrodes far from real electrodes produce broad distributions (high uncertainty). Uncertainty propagates through the full pipeline.

9.4 Uncertainty flow through the architecture



Each level captures a different source: **electrode level**—signal quality (impedance, contact, noise); **field level**—coverage gaps (no electrodes nearby); **prediction level**—phoneme ambiguity (articulatory similarity).

Part III

Architecture Design

10 Design Principles

Every architectural choice traces back to the observation function:

$$d_j^{(i)}(t) = g(\mathbf{p}_j^{(i)} + \Delta^{(i)},\ \mathbf{z}(t);\ e^{(i)}) + \epsilon_j^{(i)}(t) \tag{9}$$

Four modules, each solving one subproblem:

1. **Spatial Encoder**—Invert g: observations at known positions $\rightarrow$ latent state $\hat{\mathbf{z}}(t)$
2. **Observation Decoder**—Implement g: latent state + position $\rightarrow$ predicted observation
3. **Temporal Encoder**—Model dynamics: $\hat{\mathbf{z}}(1{:}T) \rightarrow$ temporally-coded representations

Symbol	Meaning	Architectural component
$d_j^{(i)}(t)$	Electrode observation	Input token
$\mathbf{p}_j^{(i)}$	MNI position (approximate)	MNI positional encoding
$\Delta^{(i)}$	Registration correction	Learnable 3D offset
$\mathbf{z}(t)$	Shared articulatory state	Spatial encoder output
$e^{(i)}$	Patient residual	Learnable embedding
$g(\cdot)$	Shared somatotopic map	Observation decoder
ϵ	Noise	Regularization

4. **Phoneme Decoder**—Classify trajectory: temporal representations → phoneme sequence

11 Spatial Encoder (∼100K params)

11.1 Electrode tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \to d)}_{\text{activity}} + \underbrace{\text{Linear}(\mathbf{p}_j + \Delta^{(i)} \to d)}_{\text{MNI PE}} + \underbrace{\text{Linear}(\text{row}_j, \text{col}_j \to d)}_{\text{Grid PE}} + \underbrace{e^{(i)}}_{\text{patient}} \tag{10}$$

Per-patient parameters ($3 + d = 67$ per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: learnable MNI offset (registration correction), initialized to $\mathbf{0}$.
- $e^{(i)} \in \mathbb{R}^d$: learnable patient embedding, initialized $\sim \mathcal{N}(0, 0.02)$.

Dead electrodes are excluded from the token set (not masked, just absent). Variable N_i is handled natively by attention.

11.2 Perceiver cross-attention with distance bias

$L = 16$ **virtual electrodes** with learnable MNI positions, initialized to cover the convex hull of source patient electrode positions:

$$\text{query}_i = \text{embed}_i + \text{MNI_PE}(\text{v_pos}_i) \tag{11}$$

Cross-attention with spatial locality prior:

$$\boxed{A_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d}} \; - \; \alpha \cdot \|\text{v_pos}_i - (\mathbf{p}_j + \Delta^{(i)})\|^2} \tag{12}$$

where $\alpha > 0$ is learnable (initialized 0.1), controlling spatial blur: large α = sharp attention (precise MNI trust), small α = fuzzy (robust to misalignment).

4-head attention, optional 1-layer self-attention (captures inter-zone interactions like lip-tongue co-articulation), then **MaxPool** across L virtual outputs → $s(t) \in \mathbb{R}^d$.

Component	Params	Notes
Activity embedding Linear($1 \to d$)	128	
MNI PE Linear($3 \to d$)	256	
Grid PE Linear($2 \to d$)	192	
Virtual electrode embeddings ($16 \times d$)	1,024	
Virtual MNI positions (16×3)	48	
Cross-attention QKV + output (4 heads)	∼16,384	
Self-attention (1 layer)	∼16,384	
LayerNorm ×2	256	
FFN ($d \to 4d \to d$) ×2	∼65,536	
Distance scale α	1	
Total spatial encoder	**∼100K**	
Per-patient $\Delta + e$	67/patient	

11.3 Dimensions and parameter count

12 Observation Decoder (∼25K params)

The forward observation model—the "renderer" in the multi-view analogy:

$$\boxed{\hat{d}_j(t) = \phi(\mathbf{p}_j + \Delta^{(i)},\ e^{(i)})^\top\ \psi(s(t))} \tag{13}$$

- $\phi = \mathrm{MLP}_{\mathrm{spatial}}(\mathbf{p}, e) \in \mathbb{R}^k$: spatial tuning vector—encodes what articulatory features position $\mathbf{p}$ is sensitive to.
- $\psi = \mathrm{MLP}_{\mathrm{latent}}(s) \in \mathbb{R}^k$: articulatory activation—encodes which tuning profiles are excited by the current state.
- $k = 32$: bilinear interaction dimensionality.

Interpretation: ϕ is the somatotopic map at position $\mathbf{p}$; ψ is the articulatory configuration. Their inner product gives predicted activity. This bilinear form is physically motivated by the separability of spatial tuning and articulatory dynamics (cf. MMGN, ICLR 2024).

Reprojection loss (self-supervised, no labels):

$$\mathcal{L}_{\mathrm{reproj}} = \frac{1}{N_i} \sum_{j=1}^{N_i} \left\| d_j(t) - \hat{d}_j(t) \right\|^2 \tag{14}$$

This loss simultaneously trains: the spatial encoder to produce accurate $\hat{\mathbf{z}}(t)$; the observation decoder to learn the somatotopic map g; $\Delta^{(i)}$ to correct MNI registration error; and $e^{(i)}$ to capture patient-specific characteristics. **No phoneme labels are required.**

With Beta-NLL (uncertainty-aware, Section 9):

$$\mathcal{L}^{\mathrm{UQ}}_{\mathrm{reproj}} = \frac{1}{N_i} \sum_j \frac{1}{2} \left(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2} \right) \cdot (\hat{\sigma}_j^2)^{0.5}_{\mathrm{sg}} \tag{15}$$

where $\hat{\sigma}_j^2$ is the predicted per-electrode variance and "sg" denotes stop-gradient.

Electrode masking: During training, randomly drop 20–40% of electrodes from encoder input but retain them in the reprojection loss. The model must spatially interpolate through the neural field—geometric masked reconstruction grounded in the observation model.

13 Temporal Encoder (∼70K params)

13.1 Multi-scale tokenization (experimentally validated)

$$\begin{aligned} s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=3,s=3)} T_3 \approx 67 \text{ tokens at } \sim 67\,\text{Hz} \quad \text{(fast transitions)} \\ s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=5,s=5)} T_5 \approx 40 \text{ tokens at } \sim 40\,\text{Hz} \quad \text{(syllable dynamics)} \\ s(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=10,s=10)} T_{10} \approx 20 \text{ tokens at } \sim 20\,\text{Hz} \quad \text{(phoneme-level)} \end{aligned} \tag{16}$$

All ≈127 tokens concatenated with learnable per-scale embeddings + sinusoidal temporal PE.

13.2 Temporal transformer

2 layers, 4 heads, d=64, FFN dim 256. Standard pre-norm transformer. Each token attends across all scales and times—the 67 Hz token at t=100 ms can attend to the 20 Hz token at t=800 ms.

13.3 Readout

For k-NN evaluation: z_{global} = mean_pool(all 127 tokens) $\in \mathbb{R}^d$. For per-position classification: z_{pos} = mean_pool(tokens within each phoneme window) $\in \mathbb{R}^d$ per position (windows: 0–333ms, 333–666ms, 666–1000ms).

14 Phoneme Decoder (∼5K params)

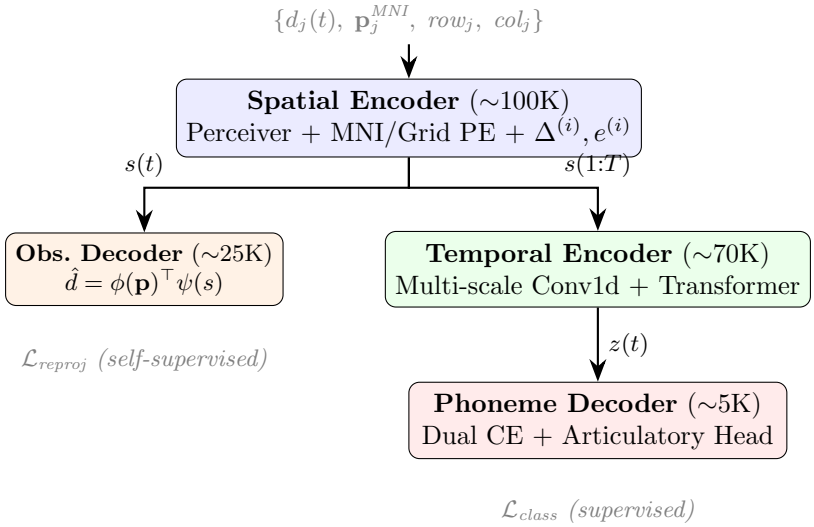
Dual head (experimentally validated):

$$\text{logits}_p = \frac{1}{2}\Big(\underbrace{\text{Linear}(z \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \text{normalize}(\text{Linear}(z \to 15)) \cdot A^\top}_{\text{articulatory head}}\Big) \tag{17}$$

where $A \in \mathbb{R}^{9\times 15}$ is the fixed articulatory feature matrix and τ is a learnable temperature.

Classification loss: $\mathcal{L}_{\text{class}} = \frac{1}{3}\sum_{p=1}^{3} \text{FocalCE}(\text{logits}_p, y_p;\ \gamma{=}2, \text{LS}{=}0.1)$ with mixup (α=0.2).

15 Complete Architecture



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \cdot \mathcal{L}_{\text{reproj}}} \tag{18}$$

Module	**Params**	**Role**
Spatial encoder	∼100K	Invert observation function
Observation decoder	∼25K	Forward model (reprojection)
Temporal encoder	∼70K	Dynamics modeling
Phoneme decoder	∼5K	Classification
Total shared	**∼200K**	
Per-patient (×10)	670	Registration + identity
Grand total	**∼201K**	(cf. current pipeline: 122K)

Part IV

Training, Evaluation, and Experimental Design

16 Training Pipeline

16.1 Phase 1: Source patient training (supervised + reprojection)

9 source patients, ∼1315 labeled trials. Each batch:

1. **Spatial encoding**: construct electrode tokens with dual PE + patient params, mask 20–40% electrodes, Perceiver cross-attention → $s(t)$ per frame.
2. **Reprojection**: observation decoder predicts all electrodes (including masked); $\mathcal{L}_{\text{reproj}}$ (or $\mathcal{L}_{\text{reproj}}^{\text{UQ}}$ with Beta-NLL).
3. **Temporal encoding**: $s(1{:}T)$ → multi-scale Conv1d → temporal transformer → $z(t)$.

4. **Classification**: dual head $\rightarrow \mathcal{L}_{\text{class}}$ with focal CE, label smoothing, mixup.
5. **Total**: $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \mathcal{L}_{\text{reproj}}$ with λ: $0.5 \rightarrow 0.1$ annealed.

Optimizer: AdamW with differential LR:

- Shared parameters: $\text{lr} = 10^{-3}$
- Virtual electrode positions: $\text{lr} = 3 \times 10^{-4}$ (slow—positions should stabilize)
- Per-patient Δ: $\text{lr} = 3 \times 10^{-4}$ (slow—geometric correction)
- Per-patient e: $\text{lr} = 10^{-3}$ (can adapt faster)

Linear warmup (20 epochs) $\rightarrow$ cosine decay. Weight decay 10^{-4}. Gradient clipping 5.0. Batch size 16. Early stopping on held-out 20% source validation, patience 7.

Optional extension: Alternate labeled batches (both losses) with unlabeled batches from continuous recordings or Lexical patients (reprojection only—no labels needed).

16.2 Phase 2: Target adaptation (LP-FT)

Per CV fold on target patient (5-fold grouped-by-token):

1. **Linear probe** (freeze encoder): Initialize $\Delta^{(\text{target})} = 0$, $e^{(\text{target})} = \text{mean}(e^{(\text{source})})$. Train phoneme decoder + Δ + e only. 150 epochs.
2. **Fine-tune** (unfreeze): Unfreeze spatial + temporal encoder at 0.1× LR. Continue all params. 100 epochs, patience 7.

16.3 Phase 3 (optional): Reprojection pre-training

Pre-train spatial encoder + observation decoder on all available data (continuous recordings + Lexical patients) using $\mathcal{L}_{\text{reproj}}$ only. No phoneme labels needed. Then initialize Phase 1 from pre-trained weights. This is the SSL component—grounded in the reprojection objective rather than arbitrary masked prediction.

17 Evaluation Protocol

Primary metric: PER via edit distance, grouped-by-token 5-fold CV.

Full evaluation recipe (all experimentally validated):

Component	Setting	Source
Weighted k-NN	k=10, cosine similarity	Autoresearch exp9
TTA	16 augmented copies	Autoresearch exp16
Dual head	CE + articulatory, averaged	Autoresearch exp14–15
Best-of per fold	$\min(\text{linear}, \text{kNN})$ per fold	Autoresearch exp9
Source k-NN	Source embeddings at 0.5× weight	LOPO exp13
Conformal prediction	Coverage-guaranteed prediction sets	Section 9
MC Dropout	10–20 passes for epistemic UQ	Section 9

ID	Ablation	Question
A1	MNI PE present/absent	**Central question**: do MNI coordinates help?
	a. Current Conv2d + BiGRU baseline	→ PER 0.762
	b. Perceiver + Grid PE only (no MNI)	→ architecture benefit
	c. Perceiver + MNI PE only (no Grid)	→ MNI benefit
	d. Perceiver + MNI + Grid PE	→ combined benefit
	e. (d) + reprojection loss	→ reprojection benefit
A2	Population-level (all patients)	Is the 0.76 wall S14-specific or universal?
A3	Per-patient params (Δ/e/both/neither)	Which per-patient parameters matter?
A4	Virtual electrode count ($L \in \{4, 8, 16, 32\}$)	How many virtual electrodes needed?
A5	Reprojection weight ($\lambda \in \{0, 0.01, 0.1, 0.5\}$)	Does reprojection help or hurt classification?
A6	Cross-task pooling (Duraivel data)	Does more data help? (architecture-independent)
A7	Uncertainty stack (Beta-NLL vs. MSE reproj)	Does learned variance improve convergence?

18 Ablation Experiments

19 Available Data and Experimental Paths

19.1 Currently used

Data source	Volume	Usage
PS task trial-level HGA	9 patients, ∼1315 trials	Supervised LOPO training
S14 trial-level HGA	153 trials	Target patient evaluation
Grid topology (from TSV)	Per-patient	SpatialConv input structure

19.2 Available but unused

Data source	Volume	Potential usage	Risk
Cross-task pooling (Duraivel)	3 patients, 156–208 trials	Doubles labeled data	Low
MNI coordinates	All patients	Spatial alignment	Medium
Continuous recordings	∼168 min	Reprojection SSL	Medium
Lexical task patients	13 patients	Additional sources	Medium
Audio recordings	4 variants/patient	Auxiliary supervision	Medium

Highest priority: Cross-task pooling (same tokens, same phonemes, overlapping patients, compatible preprocessing). **Hard dependency**: MNI coordinate sourcing from BioImage Suite / Brainlab / ECoG_Recon on Box. **Lower priority**: Continuous recording extraction + VAD curation for Phase 3 reprojection SSL. The continuous data caveat: ∼168 minutes are dominated by silence, OR ambient noise, and anesthesia fluctuations; aggressive curation required.

20 MNI Coordinate Sourcing

The architecture requires MNI-152 coordinates per electrode. Current pipeline uses ACPC-normalized grid positions (0–1 range in TSVs) for grid topology only.

Where they likely are: BioImage Suite outputs (DBS patients), Brainlab outputs (tumor patients), ECoG_Recon data on Box, BIDS coordsystem JSON files.

Fallbacks: (1) Use ACPC grid positions scaled to approximate mm—loses cross-patient alignment but retains within-patient structure. (2) Estimate MNI from grid positions + array metadata (center in MNI from surgical reports + grid geometry).

21 Risk Assessment

Risk	Severity	Mitigation
MNI coordinates unavailable	High	Fallback to scaled ACPC + Grid PE only
MNI alignment too imprecise	High	Ablation A1; learnable α adapts spatial blur
Reprojection hurts classification	Medium	λ ablation A5; can set λ=0
Perceiver overfits (200K on 1300 trials)	Medium	Reprojection regularizes; masking; early stopping
Virtual electrodes collapse	Low	Initialize across convex hull; monitor
Per-patient Δ overfits	Low	L2 penalty; low learning rate

22 Success Criteria

Level	Criterion
Minimum	A1d (Perceiver + MNI + Grid) outperforms A1b (Grid only) by $\geq$2 pp on S14
Target	PER < 0.740 on S14 (exceeds per-patient supervised 0.737)
Strong	Population-level improvement across all patients
Interpretability	Virtual electrodes cluster over vSMC; $\Delta^{(i)}$ larger for tumor patients; $\phi(\mathbf{p})$ recovers somatotopy

23 Summary

This document formalizes cross-patient speech decoding as **multi-view neural field reconstruction**: multiple patients observe the same cortical dynamics from different electrode positions, analogous to multiple cameras observing the same 3D scene.

The theoretical foundation rests on three claims of increasing specificity:

1. **Externally established**: Articulatory somatotopy exists in vSMC (Metzger, Bouchard); motor cortex manifold dynamics are preserved across individuals (Safaie, Gallego); per-patient input + shared backbone is optimal for transfer (Singh, Willett); MNI coordinates enable cross-patient ECoG decoding (Chen/SwinTW).

2. **Internally observed**: LOPO transfer works (PER 0.762); articulatory bottleneck and multi-scale temporal encoding improve performance; 55 experiments converge to PER 0.750–0.780.

3. **Working hypothesis**: The shared observation function $g(\mathbf{p}, \mathbf{z})$ factorizes the somatotopic map from articulatory dynamics, and MNI coordinates can align this function across patients.

The Neural Field Perceiver implements this framework: Perceiver cross-attention with MNI distance bias inverts the observation function ($\sim$100K), a bilinear observation decoder implements it forward for reprojection ($\sim$25K), a multi-scale temporal transformer models articulatory dynamics ($\sim$70K), and per-position dual heads decode phonemes ($\sim$5K). Three-level uncertainty estimation (Beta-NLL, MC Dropout, conformal prediction) provides per-electrode, per-position, and per-prediction

confidence. Every architectural component has a validated analog in the 2024–2026 3D vision literature.

Total budget: $\sim$200K shared parameters + 67 per patient. The architecture directly maps every component to the observation equation $d_j^{(i)}(t) = g(\mathbf{p}_j^{(i)} + \Delta^{(i)}, \mathbf{z}(t); e^{(i)}) + \epsilon_j^{(i)}(t)$.